



COLLEGE *of*
COMPUTER STUDIES

Platform Architecture

Layers of a Computing Platform

- **Hardware**
 - Physical components (CPU, memory, storage, peripherals)
- **Firmware**
 - BIOS/UEFI, device firmware, initialization processes
- **Operating System (OS)**
 - Manages hardware, runs applications, enforces security
- **Applications**
 - Software that provides user-level services

Mobile vs. Desktop Platforms

- **Mobile Platforms** (Android/iOS)
 - Optimized for portability & battery life
 - Tight integration of hardware & software
 - App sandboxing and stricter permissions (Android/iOS)
- **Desktop Platforms** (Windows, macOS, Linux)
 - Higher performance & flexibility
 - Broader software ecosystem
 - More customization, but larger attack surface

When analyzing the **platform architecture** of mobile vs. desktop systems, the distinction lies in how the hardware components, CPU instruction sets, operating system kernels, and security boundaries are layered together.

Cloud & Virtualization as Platforms

•Cloud Platforms

- Abstract physical resources, provide services on demand (Amazon Web Services (AWS), Azure)
- Security relies on shared responsibility model

•Virtualization Platforms

- Hypervisors enable multiple VMs on one physical machine
- Increases resource efficiency but adds risks (VM escape, misconfigurations)

When comparing **cloud platforms** vs. **virtualization platforms**, the fundamental difference is that virtualization is the core technology, while cloud computing is the service model built on top of it.

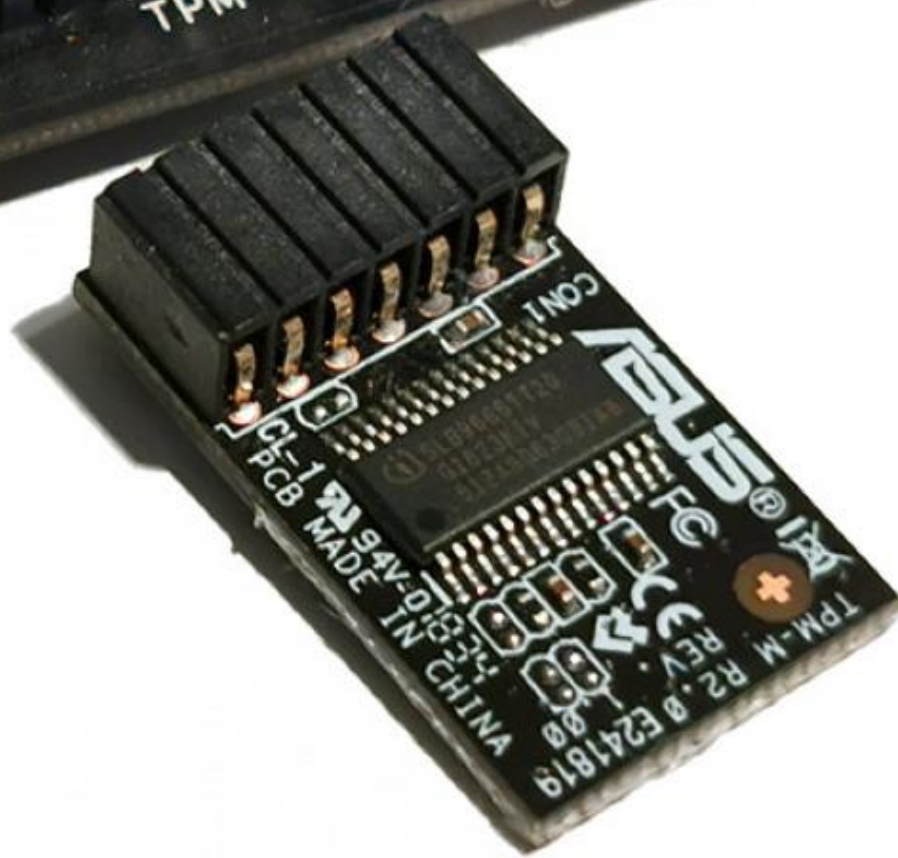
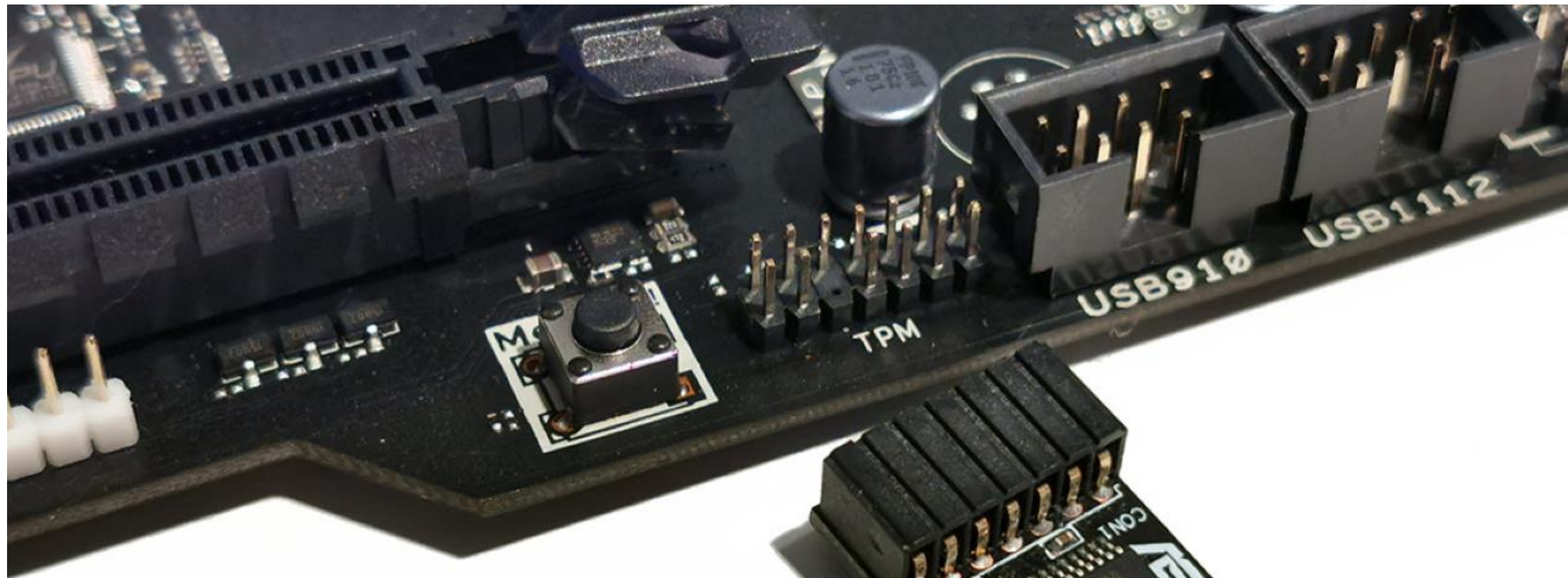
Sample Platform Attack Surface

- **Hardware:** Side-channel attacks, physical tampering (refers to an attack where an adversary gains direct, hands-on physical access to the underlying hardware hosting a platform).
- **Firmware:** BIOS/UEFI rootkits, firmware backdoors
- **OS:** Privilege escalation, kernel exploits
- **Applications:** Malware, unpatched software vulnerabilities



COLLEGE *of*
COMPUTER STUDIES

Trusted Platform Module (TPM)

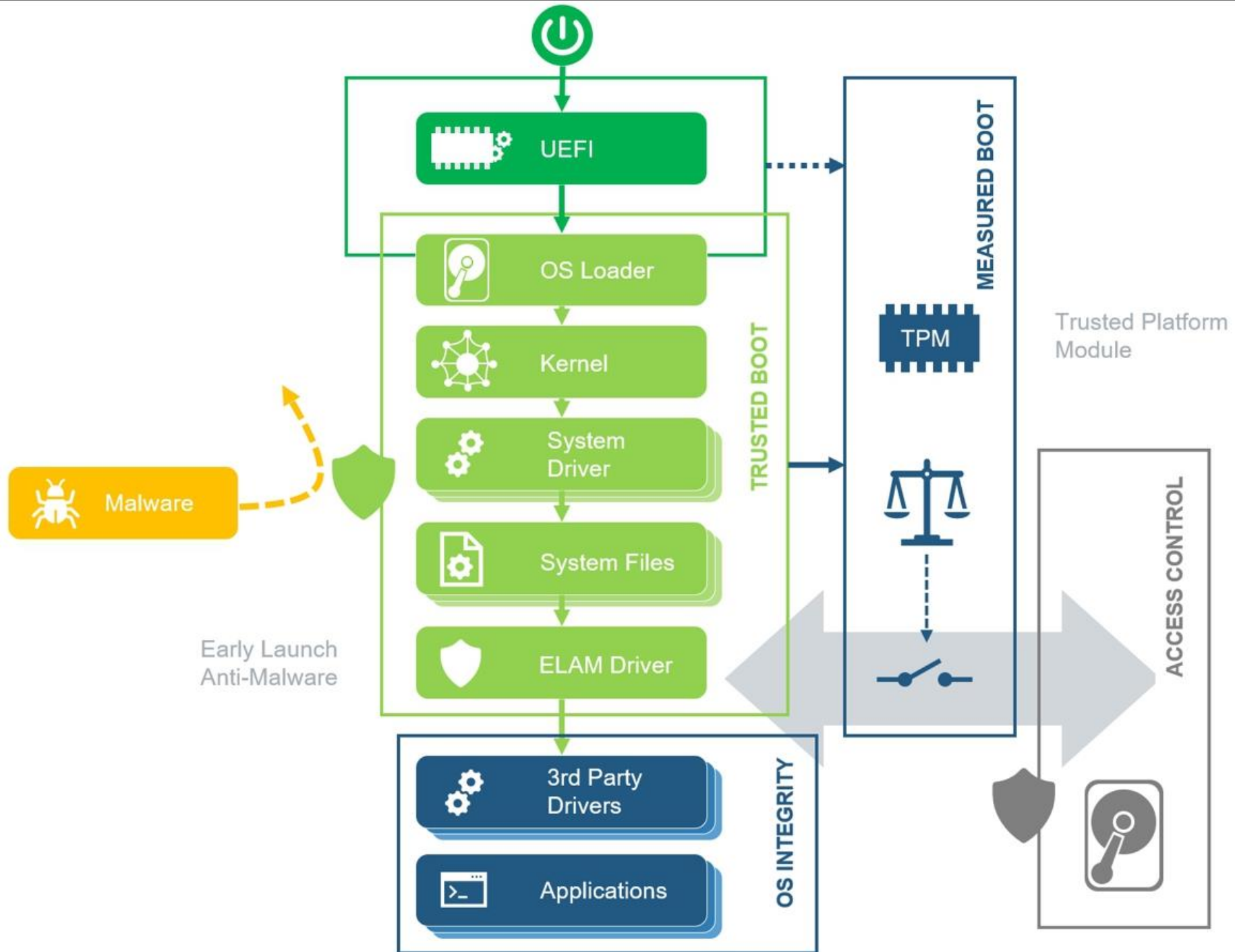


Trusted Platform Module

- TPM is a hardware-based security chip
- Provides a Root of Trust for a computing platform
- Functions independently of the main CPU and OS
- Ensures secure storage, encryption, and attestation

Why TPM Matters

- **Keeps secrets safe:**
TPM stores important “digital keys,” certificates, and passwords in a secure chip, so hackers can’t easily steal them. (Think of it like a tiny safe built into your computer.)
- **Helps your computer start safely (Secure Boot):**
When you turn on your computer, TPM checks that the system hasn’t been tampered with before letting it load.
- **Blocks hidden attacks in startup (firmware/boot-level attacks):**
Hackers sometimes try to sneak malware in the system before Windows or Linux even loads. TPM helps stop these early attacks



Unified Extensible Firmware Interface (UEFI)

- It's basically the modern replacement for BIOS (Basic Input/Output System), which is the traditional firmware that starts your computer when you power it on.
- When you press the power button, UEFI is the first program that runs.
- It initializes the hardware (CPU, RAM, drives, etc.) and then hands control over to the operating system (Windows, Linux, etc.).
- Think of UEFI as the bridge between the hardware and the OS.

Chain of Trust

- A process where each stage of the boot process checks the integrity of the next stage before handing control.
 - A **Chain of Trust** is a foundational security architecture designed to ensure that every piece of software running on a platform is verified, unmodified, and trusted, starting from the moment power hits the motherboard.
1. **Root of Trust (TPM/firmware key)** – trust anchor/immutable baseline that starts the system
 2. **Firmware checks bootloader** (must be signed/verified)
 3. **Bootloader checks OS kernel**
 4. **OS verifies drivers/applications**
- If any step fails, boot process is stopped or alerts are raised

TPM Versions

The specifications, governed by the **Trusted Computing Group (TCG)**, have two major historical eras—TPM 1.2 and TPM 2.0—along with modern micro-revisions.

- **TPM 1.2:** Basic cryptographic support (SHA-1, RSA)
- **TPM 2.0:** Stronger crypto algorithms (SHA-256, ECC), flexible authorization

TPM 2.0 is required for:

- Windows 11
- Modern enterprise-grade security solutions

Direct Comparison of TPM 1.2 vs. TPM 2.0

Feature	TPM 1.2 (Legacy Era)	TPM 2.0 (Modern Standard)
Release Timeline	Finalized between 2005–2011 .	Released in 2014 ; continuously updated.
Cryptographic Agility	None . Algorithms are hardcoded directly into the physical silicon.	High . Built as a flexible library to support multiple algorithms simultaneously.
Hashing Algorithms	Strictly limited to SHA-1 .	Supports SHA-256 , SHA-384, SM3, and higher.
Asymmetric Encryption	Natively restricted to RSA (up to 2048-bit) .	Supports RSA , Elliptic Curve Cryptography (ECC), and SM2.
Primary Boot Target	Optimized for older Legacy BIOS layouts.	Requires a modern UEFI Boot environment.

TPM Security Benefits

- Protects against:
 - Rootkits & bootkits
 - Key theft
 - Unauthorized firmware changes
- Provides **hardware-enforced trust** instead of relying on software-only security

TPM Limitations

- Requires hardware support (not all devices have it)
- If TPM fails, access to encrypted data may be lost
- Cannot stop OS-level malware once system is booted
- Physical attacks on the chip (advanced threat) still possible



COLLEGE *of*
COMPUTER STUDIES

ITA 216 Platform Security

Virtualization

Virtualization

Virtualization is a powerful technology that enables multiple virtual machines (VMs) to run concurrently on a single physical server.

Consolidated Hardware

- One physical server hosts many virtual environments.

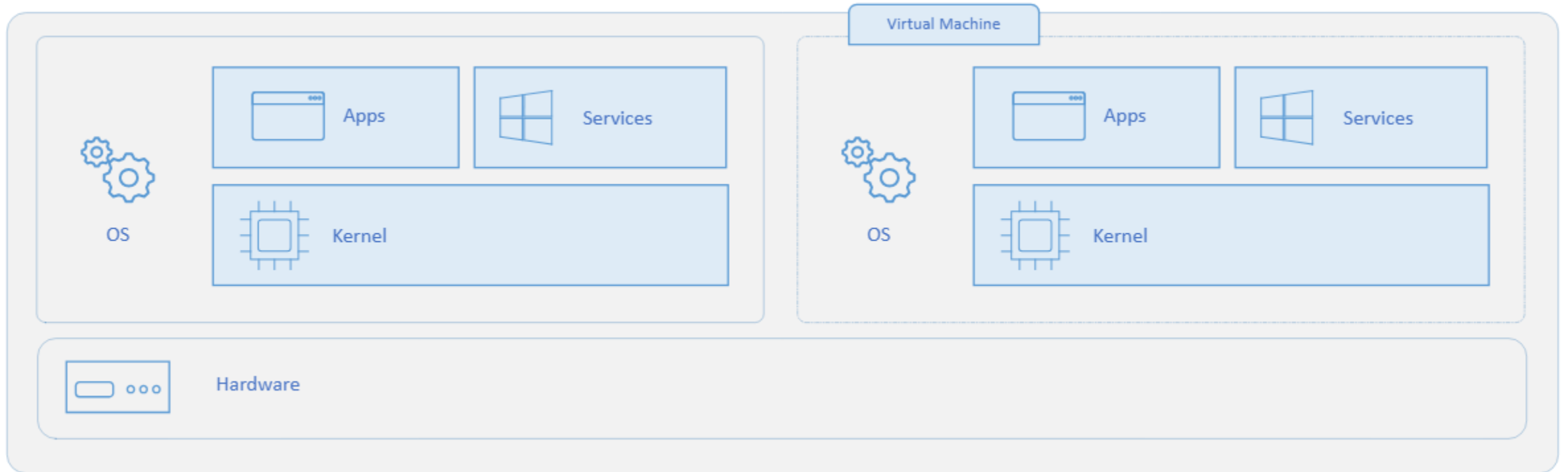
Isolated Environments

- Each VM operates independently with its own OS and applications.

Maximized Utilization

- Optimizes the use of underlying hardware resources.

Virtual Machine



Types of Virtualization

1. Server Virtualization

- Multiple virtual servers on one physical server
- Most common type

2. Desktop Virtualization

- Virtual desktop infrastructure (VDI)
- Remote desktop access

3. Network Virtualization

- Virtual networks independent of physical hardware

4. Storage Virtualization

- Pool storage from multiple devices

5. Application Virtualization

- Applications run in isolated environments

Types of Hypervisor

The core of virtualization is the **hypervisor**, a software layer that manages and allocates physical hardware resources to virtual machines, ensuring their isolation and efficient operation.

Type 1: Bare-Metal

- Runs directly on hardware, offering high performance and security (e.g., VMware ESXi, Microsoft Hyper-V).

Type 2: Hosted

- Runs on top of a host operating system (e.g., VirtualBox, VMware Workstation).

Benefits of Virtualizations

Virtualization offers significant advantages for modern businesses:

1. **Cost Reduction**-Instead of spending millions upfront on physical servers that sit idle, companies use modern platforms
2. **Increased Agility** -organization to rapidly deploy, scale, adapt, or recover its software systems with minimal friction, time, or cost.
3. **Enhanced Disaster Recovery**- restore a company's software and data instantly after a major outage or disaster.
4. **Improved Management**- Instead of an IT engineer manually configuring individual physical machines, writing command-line scripts for basic patches, or physically tracking hardware health, modern cloud and virtualization platforms consolidate everything into a single operational interface.
5. **Testing & Compatibility**- Instead of testing code on real phones, laptops, or individual server blades, developers use software-defined environments to simulate any combination of operating systems, screen sizes, or hardware requirements instantly.

Popular Virtualization Platforms

Enterprise Solutions:

- VMware vSphere
- Microsoft Hyper-V
- Citrix XenServer
- Red Hat Virtualization

Desktop/Development:

- VMware Workstation/Fusion
- Oracle VirtualBox
- Parallels Desktop
- QEMU/KVM

Security Risks of Virtualization

- Hyperjacking (compromised hypervisor)
- Escaping from VM to host
- Misconfiguration vulnerabilities

Hyperjacking (Compromised Hypervisor)

A cyberattack where the attacker gains control over the hypervisor itself, effectively taking over all hosted virtual machines.

- **Why it's dangerous:** The hypervisor has the highest privilege level — if compromised, attackers can monitor, manipulate, or shut down all VMs.
- **Real-world example:** A malicious hypervisor installed underneath an existing OS (a “blue pill” attack) to control the system invisibly.

Countermeasures:

1. Apply regular hypervisor patching.
2. Limit admin/root access with MFA and strict role separation.

Escaping from VM to Host (VM Escape)

An attack where malicious code running inside a virtual machine breaks isolation and gains access to the hypervisor or host system.

Why it's dangerous:

- Once the attacker reaches the host, they can control all other VMs.
- This violates the core promise of virtualization — isolation.

Misconfiguration Vulnerabilities

Weaknesses introduced not by flaws in the hypervisor software, but by incorrect or insecure configuration by administrators.

Examples of risky misconfigurations:

1. Assigning too many privileges to VM users (e.g., unrestricted root/admin rights).
2. Improperly configured virtual networks (e.g., flat networks without VLANs or segmentation).
3. Weak or default management console credentials.
4. Overcommitting resources (CPU, RAM) leading to denial-of-service (DoS) attacks.

VM Isolation Techniques

1. Strong Separation: Each VM Must Act as an Independent System
2. Resource Allocation Controls: CPU, RAM, Storage Quotas
3. Access Control: Prevent VM-to-VM Unauthorized Access
4. Snapshots Monitoring: Detect Rollback Attacks

Secure VM Networking

- Virtual switches: Control VM traffic
- Segmentation: VLANs for separating workloads
- Firewall rules: Per-VM or per-network segment
- IDS/IPS integration: Detect malicious VM traffic

VM Snapshots and Rollback Security

- **Snapshots:** Save system state for recovery/testing
- **Risks:**
 - Rollback to vulnerable versions
 - Exposure of sensitive data in snapshots
- **Best Practices:**
 - Encrypt snapshots
 - Monitor and control snapshot creation
 - Regularly patch and update after rollback

Best Practices for Virtualization Security

- Keep hypervisor updated and patched
- Enforce strict access controls
- Use security baselines: CIS (Center of Internet Security), NIST – (National Institute of Standards and Technology)
- Monitor VM behavior with logs & alerts
- Encrypt VM images and storage